Year 12 Further Maths - Circular motion

Similar questions - Beads, shells and pendulums

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A small bead of mass 0.2 kg is threaded on a smooth circular wire of radius 1.5 m fixed in a vertical plane with centre O. The bead is projected from the lowest point of the wire with speed 9 $\mathrm{m\,s^{-1}}$.

(a) Find the speed of the bead at the highest point of the wire. **[3]**

(b) Find the magnitude of the reaction of the wire on the bead at the lowest point. **[2]**

(c) Find the magnitude and direction of the reaction of the wire on the bead at the highest point. **[3]**

(d) Find the least projection speed from the lowest point for which the bead makes complete circles, and explain why this differs from the value for a particle on a string. **[4]**

2 A small body P of mass 2 kg is at rest at the lowest point of the inside of a smooth hemispherical shell of radius 2.5 m and centre O. P is projected horizontally with speed u $\mathrm{m\,s^{-1}}$. When P first comes to instantaneous rest, OP makes an angle of 50° with the downward vertical through O.

(a) Find u. **[4]**

(b) When OP makes an angle of 30° with the downward vertical, find the reaction of the shell on P. **[5]**

(c) State one assumption made in modelling the motion of P. **[1]**

3 A smooth wire is shaped into a circle of radius 3.6 m, fixed in a vertical plane with centre O. A small bead B is threaded on the wire. B is held so that OB makes an angle of 45° with the downward vertical through O, and is projected downwards along the wire with speed u $\mathrm{m\,s^{-1}}$. In its subsequent motion B describes complete circles about O. Given that the lowest speed of B in its motion is 5 $\mathrm{m\,s^{-1}}$, determine the value of u. **[5]**

4 A smooth hemispherical shell of radius r m is held with its circular rim horizontal and uppermost. The centre of the rim is O and the lowest point of the inner surface is A. A small object P of mass m kg is held at rest on the inner surface of the shell so that $\angle POA = \frac{1}{4}\pi$ radians. At the instant that P is released, an impulse of magnitude I N s is applied to P along the tangent at P in the vertical plane containing O, A and P, so that P moves along the surface towards the rim. X is the point where P reaches the rim. In an initial model, P experiences no resistance.

(a) Find an expression, in terms of r, g, m and I, for the speed of P when it reaches X. **[4]**

(b) Find an expression for the maximum height above X reached by P after it leaves the shell. **[2]**

(c) Find an expression for the greatest mass of P for which it still reaches X. **[2]**

In a revised model P experiences a resistive force of constant magnitude R N while it moves.

(d) Show that, in order for P still to reach X, $I > \sqrt{\sqrt{2}m^2gr + \frac{1}{2}\pi mrR}$. **[3]**

(e) Show that this inequality is dimensionally consistent. **[2]**

5 A particle P of mass 2 kg is attached to one end of a light rigid rod of length 3 m. The other end of the rod is hinged at a fixed point O and P hangs in equilibrium directly below O. A horizontal impulse of magnitude 14 N s is applied to P. In an initial model of the subsequent motion the rod is light and inextensible and resistance to the motion of P is ignored. P moves in a circular path in a vertical plane containing O. The angle that the rod makes with the downward vertical through O is θ radians.

(a) Find the largest value of θ in the subsequent motion. **[4]**

In a revised model the resistance to the motion of P is not ignored. Instead, it is modelled as causing a loss of energy of 8 J for every metre of path travelled by P.

(b) Show that, according to the revised model, the maximum value of θ satisfies $24\theta + 58.8(1 - \cos\theta) = 49$. **[3]**

(c) You are given that $\theta = 0.9722$ is a solution of this equation, correct to 4 decimal places. Determine the difference in the maximum height reached by P in the two models, giving your answer correct to 3 significant figures. **[3]**

(d) Suggest one further improvement that could be made to the model. **[1]**

Question 1

(a) Energy: $\frac{1}{2}v^2 = \frac{1}{2}(81) - g(3)$, so $v^2 = 81 - 58.8 = 22.2$ and $v = 4.71\ \text{m s}^{-1}$

(b) $R - 0.2g = \dfrac{0.2(81)}{1.5}$, so $R = 1.96 + 10.8 = 12.76$ N

(c) Towards O (downwards): $R + 0.2g = \dfrac{0.2(22.2)}{1.5} = 2.96$, so $R = 2.96 - 1.96 = 1.0$ N (directed downwards)

(d) A bead cannot go slack, so it only needs to reach the top with $v \geqslant 0$: $u^2 \geqslant 4gr = 58.8$, $u \geqslant 7.67\ \text{m s}^{-1}$. A particle on a string needs tension $\geqslant 0$ at the top, so $v^2 \geqslant gr$ there, and $u^2 \geqslant 5gr = 73.5$ ($u \geqslant 8.57$)

Question 2

(a) $\frac{1}{2}u^2 = gr(1 - \cos 50°)$: $u^2 = 2(9.8)(2.5)(0.35721) = 17.50$, $u = 4.18$

(b) Speed: $v^2 = 17.50 - 2(9.8)(2.5)(1 - \cos 30°) = 10.94$. Radially: $R - 2g\cos 30° = \dfrac{2(10.94)}{2.5}$, so $R = 16.97 + 8.75 = 25.7$ N

(c) For example: P is a particle, the shell is smooth (no friction), no air resistance

Question 3

The lowest speed occurs at the top. Height of the top above the lowest point $= 7.2$; initial height of B above the lowest point $= 3.6(1 - \cos 45°) = 1.054$. Energy: $\frac{1}{2}u^2 + g(1.054) = \frac{1}{2}(25) + g(7.2)$, so $u^2 = 25 + 2g(7.2 - 1.054) = 25 + 120.5 = 145.5$ and $u = 12.1\ \text{m s}^{-1}$

Question 4

(a) Initial speed I/m. P rises from $r(1 - \cos 45°)$ to r above A, a gain of $r\cos 45° = r/\sqrt{2}$. $v^2 = \dfrac{I^2}{m^2} - 2g\dfrac{r}{\sqrt{2}}$, so $v = \sqrt{\dfrac{I^2}{m^2} - \sqrt{2}gr}$

(b) At X the velocity is vertical, so $h = \dfrac{v^2}{2g} = \dfrac{I^2}{2gm^2} - \dfrac{r}{\sqrt{2}}$

(c) Need $v^2 > 0$: $m^2 < \dfrac{I^2}{\sqrt{2}gr}$, so $m < \dfrac{I}{\sqrt{\sqrt{2}gr}}$

(d) Arc length from P to X is $r \cdot \frac{\pi}{4}$. Work against resistance $= \frac{\pi rR}{4}$. Energy: $\frac{1}{2}mv^2 = \dfrac{I^2}{2m} - mg\dfrac{r}{\sqrt{2}} - \dfrac{\pi rR}{4} > 0$, so $I^2 > \sqrt{2}m^2gr + \frac{1}{2}\pi mrR$

(e) $[I] = \text{MLT}^{-1}$ so $[I^2] = \text{M}^2\text{L}^2\text{T}^{-2}$. $[m^2gr] = \text{M}^2 \cdot \text{LT}^{-2} \cdot \text{L} = \text{M}^2\text{L}^2\text{T}^{-2}$. $[mrR] = \text{M} \cdot \text{L} \cdot \text{MLT}^{-2} = \text{M}^2\text{L}^2\text{T}^{-2}$. All terms have the same dimensions

Question 5

(a) $I = mv$: $v = 7\ \text{m s}^{-1}$, KE $= 49$ J. Energy: $58.8(1 - \cos\theta) = 49$, i.e. $\cos\theta = \frac{1}{6}$ and $\theta = 1.40$ rad (80.4°)

(b) Distance travelled $= 3\theta$, energy lost $= 8(3\theta) = 24\theta$. Initial KE $= 49 = 2g(3)(1 - \cos\theta) + 24\theta$, so $24\theta + 58.8(1 - \cos\theta) = 49$

(c) Initial model: $h = 3(1 - \cos 1.4033) = 2.50$ m. Revised: $h = 3(1 - \cos 0.9722) = 1.31$ m. Difference $= 1.19$ m

(d) For example: include air resistance depending on speed, or allow for the mass and flexibility of the rod, or treat the resistance as varying with position